

ON FINE-TUNED BOUNDED RISK SEQUENTIAL POINT ESTIMATION OF THE MEAN OF AN EXPONENTIAL DISTRIBUTION

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Summary: The bounded risk point estimation problems for the mean of a one-parameter exponential population are addressed under the squared error loss function. Both fully sequential as well as accelerated sequential methodologies are utilized and the second-order approximations to the associated risk functions are obtained. The idea of *fine-tuning* these sampling strategies is pursued so that the associated risk functions can be expanded as the sum of the nominal preassigned bound and a remainder term which is shown to have the same $o(\cdot)$ order as that obtained by the original fully sequential process. The explicit expressions of these fine-tuning factors have been provided through our analysis.

1. INTRODUCTION

In a variety of statistical problems concerning estimation, testing, selection and ranking etc., often a purely sequential stopping rule is used that typically looks like:

$$(1.1) \quad N = N(h_0) = \inf \{n \geq m : n \geq h_0 T_n^s\}$$

where $h_0(>0)$ and $s(>0)$ are known and fixed numbers, while T_n is a positive (a.s.) statistic having some continuous distribution. Here, m is the starting sample size. In specific instances, in order to quantify the appropriateness of the stopping rule (1.1), one often is led to the evaluation of

$E[g(N/n^*)]$ where g is a sufficiently "nice" function from R^+ to R^+ and n^* is the fictitious (or unknown) "optimal" fixed-sample solution of the underlying inference problem. In some problems, one can derive an asymptotic second-order expansion that typically looks like:

$$(1.2) \quad E[g(N/n^*)] = g(1) + A^* n^{*-1} + o(n^{*-1})$$

under appropriate conditions on $g(\cdot)$, m , T_n and N . One is referred to Woodroffe (1977, 1982), Aras and Woodroffe (1993), Mukhopadhyay (1987, 1988, 1991), Ghosh and Mukhopadhyay (1989), and Isogai and Uno (1994), among others, for some of the related examples.

Usually, in the second-order expansion (1.2), A^* is a computable number. But even so, the term in the middle remains unknown since n^* is unknown in the first place. In other words, one does not really know how close $E[g(N/n^*)]$ is to its asymptotic counterpart $g(1)$, and hence the practical usefulness of expansions such as (1.2) becomes severely limited. Recently, Mukhopadhyay and Datta (1993) proposed a way to *fine-tune* the purely sequential process (1.1) that would enable one to get rid of the "problematic" middle term in (1.2).

But, the examples included in Mukhopadhyay and Datta (1993) consisted mostly of one- and multi-sample problems involving normal and two-parameter negative exponential distributions. They were able to provide a general theory in order to obtain the expression of the fine-tuning factor because in their formulation and problems, the presence of nuisance parameters made it possible to define stopping rules based on statistics T_n which were ancillary relative to the parameters of primary interest. Consequently, the conditions of theorem 2.1 of Mukhopadhyay and Datta (1993) were satisfied in all the problems handled therein.

However, in the *bounded risk point estimation* problem for the mean of a one-parameter exponential population under the *squared error loss function*, one no longer has the independence of the stopping variable and the proposed estimator at the termination stage. In other words, the sequential risk function is not expressible exclusively in the form of $E[g(N/n^*)]$ for some "nice"

function $g(\cdot)$. Yet, we are able to obtain an expansion like the one given in (1.2) by means of direct calculations along the lines of Woodroffe (1977) and Ghosh and Mukhopadhyay (1989). We should emphasize that such an expansion does not immediately follow from section 3.2 of Woodroffe (1977).

The paper is laid out as follows. Section 2 starts out with the basic bounded risk point estimation problem under a purely sequential process along the lines of (1.1). Theorem 2.1 provides an expansion such as (1.2) for the associated risk function. Then the idea of fine-tuning is introduced and theorem 2.2 gives the expression of the required fine-tuning factor. Section 3 introduces the analog of the acceleration technique of Mukhopadhyay (1993) for the present problem. The acceleration technique starts off purely sequentially and proceeds part of the way followed by batch sampling of all remaining observations in one single step. Theorems 3.1 and 3.2 respectively provide a second-order expansion of the risk function along the lines of (1.2) for the accelerated sequential rule and the expression of the required fine-tuning factor associated with the corresponding fine-tuned accelerated version. In section 4, we provide results on moderate sample performances of these procedures and give brief remarks. In practice, accelerated sequential methodologies have an edge over their purely sequential counterparts in view of substantial operational savings, and we portray a total picture by providing second-order asymptotics and foundations for all methodologies involved.

2. PURELY SEQUENTIAL SAMPLING AND ITS FINE-TUNED VERSION

Suppose that $X_1, X_2, \dots$ are independent and identically distributed (i.i.d.) random variables with the common probability density function (p.d.f.)

$$(2.1) \quad f(x; \lambda) = \lambda^{-1} \exp(-x/\lambda) I(x > 0),$$

where $\lambda \in R^+$ is the unknown mean of the distribution and $I(\cdot)$ stands for the indicator function of $(\cdot)$. Throughout, we also write $\langle u \rangle$ for the largest integer $< u$ and $u^+ = \max\{0, u\}$.

Having recorded $X_1, \dots, X_n$, consider the point estimation problem of λ by means of $\bar{X}_n \left[= n^{-1} \sum_{i=1}^n X_i \right]$, and suppose that the loss due to estimation

is measured by

$$(2.2) \quad L_n = A(\bar{X}_n - \lambda)^2,$$

where $A(>0)$ is a known constant. Such estimation problems are typically encountered in the contexts of survival analysis as well as reliability analysis. The risk associated with (2.2) is denoted by $R_n(A) = E(L_n) = A\lambda^2 n^{-1}$, which will be at most W , a preassigned positive number, provided that the sample size n is the smallest integer $\geq A\lambda^2/W = n^*$, say. This is the customary bounded risk formulation of the point estimation problem for λ .

But, since n^* is unknown, a purely sequential stopping rule is quite possibly in order along the lines of Robbins (1959), Starr and Woodroffe (1972), among others. One starts sampling with $X_1, \dots, X_m$ and then proceeds sequentially by taking one observation at a time according to the stopping rule

$$(2.3) \quad N = N(A) = \inf \{ n \geq m(\geq 1) : n \geq A\bar{X}_n^2/W \}.$$

Once the sampling terminates, λ is estimated by $\bar{X}_N$. In the following, we provide the asymptotic (as $A \rightarrow \infty$) second-order expansion of the associated risk.

THEOREM 2.1. *For the purely sequential stopping variable N in (2.3), the risk associated with $\bar{X}_N$ is given by*

$$E(L_N) = E\{A(\bar{X}_N - \lambda)^2\} = W + A^* n^{*-1} + o(n^{*-1})$$

as $A \rightarrow \infty$ if $m \geq 7$, with $D = \sum_{n=1}^{\infty} \frac{1}{2} n^{-1} E\{(\chi_{2n}^2 - 3n)^+\}$ and also

$$A^* = W(22.5 + 2D).$$

Proof: First observe that N from (2.3) can be equivalently written as

$$(2.4) \quad N = \inf \left\{ n \geq m : n^{\frac{3}{2}} n^{*- \frac{1}{2}} \geq S_n \right\},$$

where $S_n = \sum_{i=1}^n Y_i$, Y_i 's being standard exponentials with the p.d.f. $f(y; 1)$, defined in (2.1). This has the same representation as in (1.1) of Woodroffe (1977) with $\alpha = \frac{3}{2}$, $c = n^{*- \frac{1}{2}}$, $L_0 = 0$, $\mu = 1$, $\tau^2 = 1$, $\beta = 2$, $a = 1$. Let $N^* = (N - n^*)/n^{*\frac{1}{2}}$. Then, from the theorem of Ghosh and Mukhopadhyay (1975) it follows that $N^* \xrightarrow{L} N(0, 4)$ as $A \rightarrow \infty$, if $m \geq 1$. Also, lemma 2.3 and theorem 2.3 of Woodroffe (1977) lead us to conclude that

$$P\{N \leq \varepsilon n^*\} = O\left(n^{*- \frac{m}{2}}\right) \text{ for } 0 < \varepsilon < 1,$$

$$(2.5) \quad |N^*|^s \text{ is uniformly integrable, if } m > s(> 0).$$

The constant ν given in (2.4) of Woodroffe (1977) turns out to be

$$(2.6) \quad \nu = \frac{5}{4} - D,$$

and Woodroffe's (1977) theorem 2.4 leads to

$$(2.7) \quad E(N) = n^* - \frac{1}{2} - 2D + o(1) \text{ if } m > 2.$$

Let $\mathcal{R}$ denote the overshoot of the stopping variable given in (2.4), that is

$$\mathcal{R} = N^{\frac{3}{2}} n^{*- \frac{1}{2}} - S_N \text{ in other words}$$

$$(2.8) \quad (\mathcal{R} + S_N)/N = (N/n^*)^{\frac{1}{2}}.$$

From Woodroffe (1977), we claim that $\mathcal{R}^r$ is dominated for $r \geq 2$ and $E(\mathcal{R}) = \nu + o(1)$ as $A \rightarrow \infty$ if $m \geq 1$. From Anscombe (1952), it is clear that both $n^{*- \frac{1}{2}}(S_N - N)$ and $N^{- \frac{1}{2}}(S_N - N) \xrightarrow{L} N(0, 1)$ as $A \rightarrow \infty$, while from Chow et al. (1979) and (2.5), one can conclude that

$$(2.9) \quad |n^{*- \frac{1}{2}}(S_N - N)|^s \text{ is uniformly integrable if } s > 0 \text{ and } m \geq 1, \text{ and} \\ |N^{- \frac{1}{2}}(S_N - N)|^s \text{ is uniformly integrable if } s > 0 \text{ and } m > s.$$

Now, we are ready to evaluate the risk function. One writes

$$\begin{aligned}
 (2.10) \quad E[L_N] &= E[A(\bar{X}_N - \lambda)^2] = A\lambda^2 E\{(S_N - N)^2/N^2\} \\
 &= A\lambda^2 E\left\{\frac{(S_N - N)^2}{n^{*2}}\right\} + \frac{A\lambda^2}{n^{*2}} E\left\{\left[\frac{n^{*2}}{N^2} - 1\right](S_N - N)^2\right\} \\
 &= A\lambda^2 E(I) + \frac{A\lambda^2}{n^{*2}} E(J), \text{ say.}
 \end{aligned}$$

From theorem 2 of Chow et al. (1965) and (2.7), one obtains

$$(2.11) \quad E(I) = E(N/n^{*2}) = n^{*-1} - \left(\frac{1}{2} + 2D\right)n^{*-2} + o(n^{*-2})$$

if $m > 2$. Note that it will now suffice to evaluate $E(J)$ up to the $o(1)$ term. Along the lines of Ghosh and Mukhopadhyay (1989), write

$$\begin{aligned}
 (2.12) \quad E(J) &= E\left\{\left(1 - \frac{N^2}{n^{*2}}\right)(S_N - N)^2\right\} + E\left\{\left[1 - \frac{N^2}{n^{*2}}\right]^2 \frac{n^{*2}}{N^2} (S_N - N)^2\right\} \\
 &= E(J_1) + E(J_2), \text{ say.}
 \end{aligned}$$

Now,

$$\begin{aligned}
 (2.13) \quad J_2 &= \left[1 + \frac{N}{n^*}\right]^2 \left[1 + \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right]^2 \frac{n^{*2}}{N^2} \left\{\left[1 - \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right]^2 (S_N - N)^2\right\} \\
 &= a_0(N) \left\{\frac{\chi^2(S_N - N)^2}{N^2} + \frac{(S_N - N)^4}{N^2} + \frac{2\chi(S_N - N)}{N^2}\right\},
 \end{aligned}$$

in view of (2.8), with $a_0(N) = \left[1 + \frac{N}{n^*}\right]^2 \left[1 + \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right]^2 \frac{n^{*2}}{N^2}$. Consider

$$\begin{aligned}
 &a_0(N) \chi^2(S_N - N)^2 N^{-2} I(N > \tfrac{1}{2}n^*) \\
 &\leq kn^{*-1} \left[1 + \frac{N}{n^*}\right]^2 \left[1 + \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right]^2 \chi^2 \frac{(S_N - N)^2}{N} = kn^{*-1} V_N
 \end{aligned}$$

say, where $k(>0)$ is a generic constant independent of A . Since $E[V_N^{1+\varepsilon}] < \infty$ for some $\varepsilon > 0$, we obtain the uniform integrability of the lhs, that is

$E\{a_0(N)\mathcal{K}^2(S_{N^-}N)^2N^{-2}I(N > \frac{1}{2}n^*)\} = o(1)$ if $m \geq 3$. On the other hand, with $r > 1$, $s > 1$, $r^{-1} + s^{-1} = 1$,

$$\begin{aligned}
 & E\{a_0(N)\mathcal{K}^2(S_{N^-}N)^2N^{-2}I(N \leq \frac{1}{2}n^*)\} \\
 & \leq kn^{*3}E\left[\mathcal{K}^2\frac{(S_{N^-}N)^2}{n^*}I(N \leq \frac{1}{2}n^*)\right] \\
 & \leq kn^{*3}E^{\frac{1}{r}}\left[\left\{\mathcal{K}^2\frac{(S_{N^-}N)^2}{n^*}\right\}^r\right]P^s(N \leq \frac{1}{2}n^*) \\
 & = O\left[n^{*3-\frac{m}{2s}}\right] = o(1) \text{ if } m > 6s, \text{ that is if } m \geq 7,
 \end{aligned}$$

in view of Holder's inequality and (2.5), (2.9). Hence

$$(2.14) \quad E\left[a_0(N)\frac{\mathcal{K}^2(S_{N^-}N)^2}{N^2}\right] = o(1) \text{ if } m \geq 7.$$

Also, $a_0(N)\frac{(S_{N^-}N)^4}{N^2}I(N > \frac{1}{2}n^*) \leq k\left(1+\frac{N}{n^*}\right)\left[1+\left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right]^2\frac{(S_{N^-}N)^4}{N^2} = V'_N$ say.

Since $E[V_N'^{1+\varepsilon}] < \infty$ for some $\varepsilon > 0$, we obtain the uniform integrability

of the lhs, that is $E\left[a_0(N)\frac{(S_{N^-}N)^4}{N^2}I(N > \frac{1}{2}n^*)\right] = (1+1)^2(1+1)^2(1)(3) +$

$o(1) = 48 + o(1)$ if $m \geq 5$. Again as before, $E\left[a_0(N)\frac{(S_{N^-}N)^4}{N^2}I(N \leq \frac{1}{2}n^*)\right] = o(1)$ if $m \geq 5$, that is

$$(2.15) \quad E\left[a_0(N)\frac{(S_{N^-}N)^4}{N^2}\right] = 48 + o(1) \text{ if } m \geq 5.$$

These follow since $V'_N \stackrel{L}{\leq} k\chi_1^4$, if $m \geq 1$, on top of everything else. Now,

$E\left[a_0(N)\frac{\mathfrak{X}}{N^{\frac{3}{2}}}\frac{|S_{N^-}N|}{N^{\frac{1}{2}}}I\left(N > \frac{1}{2}n^*\right)\right] = o(1)$ if $m \geq 2$, but one also notes that
 $E\left[a_0(N)\frac{\mathfrak{X}}{N^{\frac{3}{2}}}\frac{|S_{N^-}N|}{N^{\frac{1}{2}}}I\left(N \leq \frac{1}{2}n^*\right)\right] \leq kn^{*2}E\left[\mathfrak{X}\frac{|S_{N^-}N|}{N^{\frac{1}{2}}}I\left(N \leq \frac{1}{2}n^*\right)\right] = o(1)$, if
 $m \geq 6$, via Holder's inequality and (2.5). That is,

$$(2.16) \quad E\left[a_0(N)\frac{\mathfrak{X}(S_{N^-}N)}{N^2}\right] = o(1) \text{ if } m \geq 6.$$

Hence, combine (2.13) with (2.14)–(2.16) to claim that

$$(2.17) \quad E(J_2) = 48 + o(1) \text{ if } m \geq 7.$$

Also, with $b(N) = -\left(1 + \frac{N}{n^*}\right)\left[1 + \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right]$, one writes

$$(2.18) \quad \begin{aligned} E(J_1) &= E\left\{b(N)\frac{\mathfrak{X}(S_{N^-}N)^2}{N}\right\} + E\left\{b(N)\frac{(S_{N^-}N)^3}{N}\right\} \\ &= E\{b(N)J_{11}\} + E\{b(N)J_{12}\}, \text{ say.} \end{aligned}$$

Along the lines of lemma 2.2 of Ghosh and Mukhopadhyay (1989), via Holder's inequality, we claim that

$$(2.19) \quad \begin{aligned} E\left\{b(N)\frac{\mathfrak{X}(S_{N^-}N)^2}{N}\right\} &= -4\nu + o(1) \\ &= -(5 - 4D) + o(1) \text{ if } m \geq 5. \end{aligned}$$

Next, observe that

$$(2.20) \quad E\{b(N)J_{12}\} = -E\left\{\left[1 + \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right](2J_{121} + J_{122})\right\},$$

where $J_{121} = (S_{N^-}N)^3/n^*$ and $J_{122} = \frac{(n^*-N)}{n^*N}(S_{N^-}N)^3$.

From theorem 9 of Chow et al. (1965), it follows that

$$(2.21) \quad E(J_{121}) = E\{2N + 3N(S_N - N)/n^*\} \\ = 2 - (1 + 4D)n^{*-1} + o(n^{*-1}) + E\{3N(S_N - N)/n^*\}$$

if $m > 2$, in view of (2.7). Also, one writes

$$3\frac{N}{n^*}(S_N - N) = 3\left[(S_N - N) + \left[1 + \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right](S_N - N)\left[-1 + \frac{N + S_N}{N}\right]\right],$$

which implies that (since $E(S_N - N) = 0$)

$$(2.22) \quad E\left[3\frac{N}{n^*}(S_N - N)\right] \\ = 3E\left\{\left[1 + \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right]\frac{(S_N - N)^2}{N}\right\} + 3E\left\{\left[1 + \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right]\frac{N(S_N - N)}{N}\right\} \\ = 6 + o(1), \text{ if } m \geq 3.$$

From (2.21)–(2.22), one gets

$$(2.23) \quad E(J_{121}) = 8 + o(1), \text{ if } m \geq 3.$$

Again, write

$$J_{122} = -\left[1 + \left[\frac{N}{n^*}\right]^{\frac{1}{2}}\right]\left\{\frac{N(S_N - N)^3}{N^2} + \frac{(S_N - N)^4}{N^2}\right\},$$

and hence one can verify that

$$(2.24) \quad E(J_{122}) = -6 + o(1), \text{ if } m \geq 5.$$

Combine (2.20), (2.23) and (2.24), and use the techniques (Holder's inequality, etc.) of Ghosh and Mukhopadhyay (1989) to conclude that

$$(2.25) \quad E\{b(N)J_{12}\} = -2\{2(8) - 6\} + o(1) \\ = -20 + o(1), \text{ if } m \geq 5.$$

Now, one combines (2.10), (2.17)–(2.19), and (2.23)–(2.25) for $m \geq 7$ to claim that

$$\begin{aligned}
(2.26) \quad E(L_N) &= Wn^* \left\{ n^{*-1} - \left(\frac{1}{2} + 2D \right) n^{*-2} + o(n^{*-2}) \right\} \\
&\quad + Wn^* n^{*-2} \{-25 + 4D + 48 + o(1)\} \\
&= W + Wn^{*-1} (22.5 + 2D) + o(n^{*-1}).
\end{aligned}$$

This completes the proof. ■

Even though the analysis given here is similar in spirit to that of Ghosh and Mukhopadhyay (1989), we provide details in the proof of theorem 2.1 for two reasons. The decompositions of the terms involved in the present analysis are not exactly the same as those in Ghosh and Mukhopadhyay (1989), and yet the evaluation of each single term is truly essential in order to come up with the required condition on the starting sample size. This is one reason for providing the details. The other reason is that in the following sections, this theorem is used as a springboard and similar types of calculations are entirely omitted from the proofs of all the theorems that follow.

2.1 THE FINE-TUNING ASPECTS

The expansion of $E(L_N)$ provided in theorem 2.1 is indeed of the form given in (1.2). Now along the lines of Mukhopadhyay and Datta (1993), we wish to propose a fine-tuned version of the purely sequential stopping rule (2.3) so that we can claim that the risk associated with the corresponding estimator of λ is actually $W + o(n^{*-1})$ if $m \geq 7$.

Let us first fix a number $\varepsilon \in R$. A very specific choice of ε will be provided in theorem 2.2. Define

$$(2.27) \quad N = N(A) = \inf \{ n \geq m : n + \varepsilon \geq A \bar{X}_n^2 / W \},$$

and once this purely sequential sampling process terminates, λ is estimated by the associated $\bar{X}_N$.

THEOREM 2.2. *For the fine-tuned purely sequential stopping variable N in (2.27), the risk associated with $\bar{X}_N$ is given by*

$$E(L_N) = E \left\{ A \left[\bar{X}_N - \lambda \right]^2 \right\} = W + o(n^{*-1})$$

as $A \rightarrow \infty$ if $m \geq 7$, with $D = \sum_{n=1}^{\infty} \frac{1}{2} n^{-1} E\left\{(\chi_{2n}^2 - 3n)^+\right\}$ and $\varepsilon = -(22.5 + 2D)$.

Proof. The stopping variable N can be rewritten in the form of (1.1) in Woodroffe (1977) with $\alpha = \frac{3}{2}$, $c = n^{*-1/2}$, $L_0 = \frac{1}{2}\varepsilon$, $\mu = 1$, $\tau^2 = 1$, $\beta = 2$ and the overshoot $\mathfrak{X} = N\left[\frac{N+\varepsilon}{n^*}\right]^{\frac{1}{2}} - S_N$ where S_n is the same as in (2.4).

Equation (2.7) is now replaced by (as $A \rightarrow \infty$)

$$(2.28) \quad E(N) = n^* - \frac{1}{2} - 2D - \varepsilon + o(1) \text{ if } m > 2.$$

Next, express $E(L_N)$ in the form of (2.10), and obtain

$$(2.29) \quad E(I) = n^{*-1} - \left(\frac{1}{2} + 2D + \varepsilon\right)n^{*-2} + o(n^{*-2}) \text{ if } m > 2.$$

Then, break up J into J_1, J_2 as in (2.12), and rewrite

$$(2.30) \quad J_2 = \left[\left[1 + \frac{N}{n^*}\right]^2 \left[1 - \frac{N+\varepsilon}{n^*}\right]^2 \frac{n^{*2}}{N^2} + \left[1 + \frac{N}{n^*}\right]^2 \frac{\varepsilon^2}{N^2} + 2\varepsilon \left[1 + \frac{N}{n^*}\right]^2 \left[1 - \frac{N+\varepsilon}{n^*}\right] \frac{n^*}{N^2}\right] (S_N - N)^2,$$

and each term is evaluated as in (2.14) – (2.16) to provide

$$(2.31) \quad E(J_2) = 48 + o(1) \text{ if } m \geq 7.$$

Also, the term J_1 is decomposed first as

$$(2.32) \quad J_1 = \left[\left[1 + \frac{N}{n^*}\right] \left[1 + \left[\frac{N+\varepsilon}{n^*}\right]^{\frac{1}{2}}\right] \left[1 - \frac{\mathfrak{X} + S_N}{N}\right] + \varepsilon \left[1 + \frac{N}{n^*}\right] n^{*-1}\right] (S_N - N)^2,$$

and then follow as in (2.18) – (2.25) to obtain

$$(2.33) \quad E(J_1) = 2\varepsilon - 25 + 4D + o(1) \text{ if } m \geq 5.$$

Now, combine (2.29) – (2.33) to write

$$\begin{aligned}
 (2.34) \quad E(L_N) &= A\lambda^2 E(I) + A\lambda^2 n^{*-2} E(J) \\
 &= Wn^* \{n^{*-1} - (\frac{1}{2} + 2D + \epsilon)n^{*-2} + o(n^{*-2})\} \\
 &\quad + Wn^* n^{*-2} \{2\epsilon - 25 + 4D + o(1) + 48 + o(1)\} \\
 &= W + Wn^{*-1} \{22.5 + 2D + \epsilon\} + o(n^{*-1}) \\
 &= W + o(n^{*-1}),
 \end{aligned}$$

if $m \geq 7$ and $\epsilon = -22.5 - 2D$. ■

3. ACCELERATED SEQUENTIAL SAMPLING AND ITS FINE-TUNED VERSION

Instead of purely sequential sampling essentially up to n^* , let us first proceed sequentially up to a fraction of n^* , followed by a single batch sampling. Hall (1983) introduced such sampling strategies for constructing a fixed-width confidence interval for the mean of a normal distribution whose variance is also unknown, because these sampling schemes are operationally more convenient than their purely sequential counterparts. A unified theory of accelerated sequential sampling was given in Mukhopadhyay and Solanky (1991). Recently, Mukhopadhyay (1993) has introduced a new and very powerful way to accelerate purely sequential schemes that provide the asymptotic second-order expansion of the associated regret function in the exponential case having a form that is similar to (3.6) in Woodroffe (1977). Our goal is to accelerate both (2.3) and (2.27), and obtain second-order expansions of the corresponding risk functions, $E(L_N)$.

3.1 Accelerated sequential sampling

Here, we address the purely sequential process (2.3). Along the lines of Mukhopadhyay (1993), first choose and fix $\rho \in (0, 1)$ such that ρ^{-1} is an integer. Define

$$(3.1) \quad t = t(A) = \inf \{n \geq m : n \geq \rho A \bar{X}_n^2 / W\},$$

a purely sequential stopping time, that is implemented in a way similar to (2.3). Observe that t approximates (or estimates) ρn^* , a fraction of n^* .

At this step, one obtains t as well as $X_1, \dots, X_t$, and lets

$$(3.2) \quad N = N(A) = \rho^{-1}t,$$

and then one samples the difference $(N - t)$ in *one single batch*. Thus, finally one obtains N and $X_1, \dots, X_t, \dots, X_N$ and estimates λ by means of $\bar{X}_N$. It is easy to see that the accelerated methodology (3.1)–(3.2) reduces the sampling operation by about $100(1-\rho)\%$ when compared with the full sequential process (2.3).

THEOREM 3.1. *For the accelerated sequential stopping variable N in (3.2), the risk associated with $\bar{X}_N$ is given by*

$$E(L_N) = E\{A(\bar{X}_N - \lambda)^2\} = W + A^{**}n^{*-1} + o(n^{*-1})$$

as $A \rightarrow \infty$ if $m \geq 7$, with $A^{**} = \{18 + \rho^{-1}(4.5 + 2D)\}W$ where

$$D = \sum_{n=1}^{\infty} \frac{1}{2} n^{-1} E\{(\chi_{2n}^2 - 3n)^+\}.$$

Proof. Along the lines of (4.4) in Mukhopadhyay (1993), one first writes

$$(3.3) \quad E(L_N) = A\rho^2 E\{(\bar{X}_t - \lambda)^2\} + A\lambda^2 \rho(1 - \rho)E(t^{-1}).$$

Under the pseudo loss function $L_n^* = A\rho(\bar{X}_n - \lambda)^2$, if the goal is to make the associated risk $E(L_n^*) \leq W$, the corresponding “optimal” fixed sample size turns out to be $n_0 = \rho n^*$, while the corresponding full sequential process will then coincide with our t in (3.1). Hence theorem 2.1 immediately provides the following expansion:

$$(3.4) \quad E\{L_t^*\} = W + W(22.5 + 2D)n_0^{-1} + o(n_0^{-1}),$$

if $m \geq 7$. Also, from theorem 3 of Mukhopadhyay (1988), one obtains

$$(3.5) \quad E\{n_0/t\} = 1 + (2D + 4.5)n_0^{-1} + o(n_0^{-1}),$$

if $m \geq 5$. Now, with $m \geq 7$, one combines (3.3)–(3.5) to write (since $n^* = \rho^{-1}n_0$),

$$\begin{aligned}
(3.6) \quad E(L_N) &= \rho \{ W + W(22.5 + 2D)n_0^{-1} + o(n_0^{-1}) \} \\
&\quad + Wn^* \rho(1 - \rho)n_0^{-1} \{ 1 + (4.5 + 2D)n_0^{-1} + o(n_0^{-1}) \} \\
&= W + n^{*-1} \{ W(22.5 + 2D) + (1 - \rho)\rho^{-1}(4.5 + 2D)W \} + o(n^{*-1}) \\
&= W + n^{*-1} \{ 18 + \rho^{-1}(4.5 + 2D) \} W + o(n^{*-1}).
\end{aligned}$$

This completes the proof. ■

One should note that the middle term of the risk-expansion given in theorem 3.1 is "little" larger than that provided in theorem 2.1. But, then, the accelerated methodology (3.1)–(3.2) also saves a significant amount of sampling operations compared with the full sequential process. In practice, the experimenter will need to balance these two facts. In many different types of related problems, we have noted through simulations that even with $\rho = \frac{1}{2}$ or $\frac{1}{3}$, the risk increment compared with the fully sequential scheme can be fairly negligible for moderate values of $n^* (\geq 100)$, while operational savings ($\approx 100(1 - \rho)\%$) can indeed be quite substantial via acceleration. In this context, compare our tables 1 and 2.

3.2. The fine-tuning aspects

Now, we combine the ideas from sections 2.1 and 3.1. Let us first fix $\varepsilon \in R$ and $\rho \in (0, 1)$, such that ρ^{-1} is an integer. The specific choice of ε is given in theorem 3.2. Define

$$(3.7) \quad t = t(A) = \inf \{ n \geq m : n + \varepsilon \geq \rho A \bar{X}_n^2 / W \},$$

$$(3.8) \quad N = N(A) = \rho^{-1}t.$$

That is, first obtain $X_1, \dots, X_t$ purely sequentially, and then sample $(N - t)$ in one single batch to finally come up with $X_1, \dots, X_t, \dots, X_N$. One proposes to estimate λ by means of $\bar{X}_N$ at this step. The following result is stated without proof for brevity.

THEOREM 3.2. *For the fine-tuned accelerated sequential stopping variable N in (3.8), the risk associated with $\bar{X}_N$ is given by*

$$E(L_N) = E\{A(\bar{X}_N - \lambda)^2\} = W + o(n^{*-1})$$

as $A \rightarrow \infty$ if $m \geq 7$, with $D = \sum_{n=1}^{\infty} \frac{1}{2} n^{-1} E\{(\chi_{2n}^2 - 3n)^+\}$ and $\varepsilon = -(18\rho + 2D + 4.5)$.

Remark 1. It is to be noted that the methodologies we presented in sections 2 and 3 as well as the techniques we used therein did not make full use of the characteristics of an exponential distribution. The only properties of this particular distribution that we exploited there are as follows: it is a scale-family of distributions where the scale parameter happens to be the mean, while the variance turns out to be the square of the mean. So, our methodologies are potentially generalizable to similar estimation-problems concerning some other members of the natural exponential family with the quadratic variance function. For brevity, we focus our attention only on the exponential distribution.

4. MODERATE SAMPLE PERFORMANCES

In this section, we present the results of the simulated experiments that we conducted to examine how the effect of "fine-tuning" is manifested in practice. For implementing the purely sequential stopping rule of section 2 and its fine-tuned version, we first generated $Exp(\lambda)$ random deviates by means of DRNEXP subroutine from IMSL (1989, Version 1.1) with $\lambda = 5$. We fixed $A = 5$ and $n^* = 100, 200, 250, 325$ and 375 , obtaining the corresponding values of W . Table 1 provides the values of $\bar{n}$, $\bar{L}$ which are estimates of $E(N)$ and $E(L_N)$ respectively, obtained from 15 000 independent replications. The estimated standard errors, namely $s(\bar{n})$ and $s(\bar{L})$ are also included. Fortran programs were written and executed on the IBM 386 main-frame computer at the University of Connecticut. We also provide the values of D , $f(D) = -0.5 - 2D$ and $g(D) = (22.5 + 2D)W/n^*$ for reference purposes.

TABLE 1. Purely sequential procedure and its fine-tuned version

 $D = 0,7117564$, $f(D) = -1,9235$, $m = 10$, $A = 5$, $\lambda = 5$ and $\epsilon = -22,5 - 2D$

W n^* $g(D)$	$\bar{n}$ $(\bar{n})^{**}$	$s(\bar{n})$ $(s(\bar{n}))$	$\bar{L}$ $(\bar{L})$	$s(\bar{L})$ $(s(\bar{L}))$	$\bar{L} - W$ $(\bar{L} - W)$
1,25 100 0,29904	97,447 (122,313)	0,665 (0,573)	1,850 (1,091)	0,092 (0,045)	0,60017 (-0,15904)
0,625 200 0,07476	197,916 (221,809)	1,019 (0,845)	0,701 (0,605)	0,035 (0,025)	0,07582 (-0,02040)
0,500 250 0,04785	247,986 (271,548)	1,013 (0,967)	0,547 (0,495)	0,023 (0,021)	0,04710 (-0,00549)
0,3846 325 0,02831	323,118 (346,265)	1,100 (1,094)	0,404 (0,376)	0,016 (0,015)	0,01936 (-0,00812)
0,3333 375 0,02127	372,631 (397,189)	1,252 (1,295)	0,352 (0,324)	0,015 (0,015)	0,01906 (-0,00923)

** The numbers in parentheses () refer to the fine-tuned version.

In the accelerated sequential scenario of section 3, we go through simulation exercises in a similar fashion with $\rho = \frac{1}{2}$. The estimates provided in table 2 were again obtained from 15 000 independent replications for each row. Again we provide the values of D , $f^*(D) = 2f(D)$ and $g^*(D) = \{18 + 2(4.5 + 2D)\}W/n^*$ for reference purposes.

From table 1, it is clear that $\bar{n}$ underestimates n^* in the purely sequential case, while $(\bar{L} - W)$ values are consistently positive. In fact, $(\bar{n} - n^*)$ values compare quite favourably with the asymptotic expression $f(D)$, whereas $(\bar{L} - W)$ values have a reasonable match with the corresponding asymptotic expression $g(D)$. For the fine-tuned version, in the

TABLE 2. Accelerated sequential procedure and its fine-tuned version

 $D = 0,7117564$, $f^*(D) = -3,8470$, $m = 10$, $A = 5$, $\lambda = 5$, $\rho = 0,5$ and $\varepsilon = -13,5 - 2D$

W n^* $g(D)$	$\bar{n}$ $(\bar{n})^{**}$	$s(\bar{n})$ $(s(\bar{n}))$	$\bar{L}$ $(\bar{L})$	$s(\bar{L})$ $(s(\bar{L}))$	$\bar{L} - W$ $(\bar{L} - W)$
1,25 100 0,37309	94,171 (127,750)	1,030 (0,827)	1,893 (1,059)	0,119 (0,054)	0,64294 (-0,19139)
0,625 200 0,09327	194,437 (227,334)	1,441 (1,281)	0,810 (0,589)	0,057 (0,030)	0,18514 (-0,03603)
0,500 250 0,05969	245,038 (276,237)	1,582 (1,448)	0,590 (0,478)	0,039 (0,024)	0,09031 (-0,02154)
0,3846 325 0,03532	320,487 (351,057)	1,765 (1,693)	0,422 (0,381)	0,026 (0,019)	0,03730 (-0,00398)
0,3333 375 0,02653	369,608 (401,260)	1,797 (1,806)	0,364 (0,331)	0,019 (0,016)	0,03044 (-0,00210)

** The numbers in parentheses () refer to the fine-tuned version.

sequential case, however, $(\bar{n} - n^*)$ values compare very favourably with the asymptotic value 22.0 (see equation (2.28)) throughout the table, whereas the $(\bar{L} - W)$ values are all seen to be small negative numbers. In other words, the proposed fine-tuning does its job even for moderate values of n^* .

In the purely sequential case, since $(\bar{L} - W)$ values are all positive for different values of n^* , the fine-tuning idea adjusts the $\bar{n}$ values upwards in order to lower the risk and brings it down under W , the target value. We also implemented the sequential rule (2.27) with $\varepsilon = 15(1)23, 23.9235$ via simulation with 15 000 replications each, and we found that $(\bar{L} - W)$ values were frequently positive and sometimes these oscillated between small

positive and negative numbers when $\varepsilon = 15(1)23$ and $n^* = 325, 375$. But, for $\varepsilon = 23.9235 (= -22.5 - 2D)$, different simulation exercises based on 15 000 replications provided negative values of $(\bar{L} - W)$ for $n^* = 325, 375$ in each and every occasion.

The entries in table 2 have similar interpretations. In the accelerated sequential case, $(\bar{n} - n^*)$ and $(\bar{L} - W)$ values are respectively compared with $f^*(D)$ and $g^*(D)$, the asymptotic values, and we find a good match there for a wide range of values of n^* . One should also realize that we achieve slightly higher positive values of $(\bar{L} - W)$ in the accelerated sequential case (with $\rho = 0.5$), but at the same time it requires approximately 50% of the sampling operations when compared with the fully sequential procedure's performances (table 1). The performances of the fine-tuned accelerated sequential version presented in table 2 are to be interpreted in the same spirit.

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